

Recap. In a second-price auction, bidding your true value is a dominant strategy: it maximizes your utility against every profile of other bids. We now ask how to allocate several advertising slots and set payments that retain this guarantee.

Position Auctions

There are k advertising positions (slots) and $n > k$ advertisers (bidders). Each advertiser can receive at most one slot, and each slot can hold at most one advertiser. Slot j has click-through rate (CTR) α_j , the probability of a click, where $1 \geq \alpha_1 \geq \dots \geq \alpha_k > 0$ and $\alpha_j = 0$ for $j > k$. Bidder i has value v_i per click. In slot j at price p per click, their expected utility is

$$u_i = \underbrace{\alpha_j}_{\text{CTR utility per click}} \underbrace{(v_i - p)}_{\text{}} .$$

Values and bids are nonnegative, and ties are broken by a fixed rule. CTRs depend only on the slot. This is the position-auction model [Edelman–Ostrovsky–Schwarz '07; Varian '07].

What is welfare?

Claim 1. Assigning slots in decreasing order of value maximizes welfare.

Proof.

The Mechanism. Rank bids $b_{(1)} \geq \dots \geq b_{(n)}$ and assign slots in that order. Charge each winner the loss in other bidders' total reported value caused by their presence. To calculate it, remove them and let the bidders below them move up one slot.

Derive the expected payment P_j for the bidder in slot j , and their price per click $p_j = P_j/\alpha_j$.

Why is $p_j \leq b_{(j+1)}$? What happens with one slot?

One advertiser's externality

Let $\alpha = (1, 0.5, 0.2, 0)$ and $v = b = (10, 8, 4, 2)$. For bidder 1, compute the other bidders' total expected value with bidder 1 present and their maximum total expected value without bidder 1. What is their expected payment?

VCG Beyond Position Auctions

Position auctions are single-parameter: one value per click determines a bidder's value for every slot. With different items, a bidder may need several numbers to describe their values. Such preferences are *multidimensional*; the single-parameter version of Myerson's lemma does not directly apply.

Let Ω be a fixed finite set of feasible outcomes. Bidder i has private value $v_i(\omega)$ for outcome ω , reports $b_i(\omega)$, and has utility $v_i(\omega) - p_i$ when charged total payment p_i . An outcome specifies the whole allocation.

The Vickrey–Clarke–Groves (VCG) mechanism chooses an outcome maximizing reported welfare (breaking ties by a fixed rule):

$$\omega^* \in \operatorname{argmax}_{\omega \in \Omega} \sum_h b_h(\omega).$$

Let W_{-i}^* be the others' maximum reported value when i 's value is omitted. Charge i the difference between this benchmark and others' reported value in the chosen outcome:

$$W_{-i}^* = \max_{\omega \in \Omega} \sum_{h \neq i} b_h(\omega), \quad p_i = W_{-i}^* - \sum_{h \neq i} b_h(\omega^*).$$

In our allocation examples, W_{-i}^* is others' optimal welfare without i : items may remain unallocated and bidders value only their own bundles.

Theorem 1. *For private values and quasilinear utilities, VCG with exact reported-welfare maximization over a fixed finite outcome set is DSIC. Under truthful reports, it maximizes welfare.*

Proof. Fix i 's true valuation and the other reports. Substitute the payment into their utility. Which term is independent of their report, and why does truthful reporting maximize the rest?

Back to slots. Why does this prove truthfulness for the position auction?

Generalized Second Price

GSP also ranks bids, but charges the bidder in slot j the next bid $b_{(j+1)}$ per click. With one slot this is second price. With multiple slots, truthful bidding need not be dominant [Edelman–Ostrovsky–Schwarz '07; Varian '07].

Exercise. There are two slots, $\alpha = (1, 0.995)$, and three bidders with $v = (10, 4, 2)$. Fix $b_2 = 4, b_3 = 2$. Compare bidder 1's utility from $b_1 = 10$ and $b_1 = 3$ under GSP. Repeat under VCG.

Does failure of DSIC imply that a game has no Nash equilibrium?